

CO3-AT1-Assignment

View Only

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QUESTIONS

Q1

Banking Transaction System –
Exception Handling

10 marks



Q2

Hospital Patient Registration –
User-Defined Exception

10 marks



Q3

E-Commerce Order Processing
– Built-in Exceptions

10 marks



Q4

Online Food Delivery – Multiple
Threads and Execution Order

10 marks



Q5

Manufacturing Plant –
Producer-Consumer Problem

Q1 Banking Transaction System – Exception Handling

10 marks

Develop a Java program for an ATM withdrawal system. The program accepts the account balance and withdrawal amount.

The program must:

- Accept the current account balance and withdrawal amount.
- Throw a user-defined `InsufficientBalanceException` when the withdrawal amount exceeds the balance.
- Handle withdrawal amounts that are zero or negative.
- Handle invalid numeric input using a suitable built-in exception.
- Display a meaningful transaction result.
- Use try-catch-finally so that the transaction completion message is always displayed.

Sample Test Cases:

Input / Scenario Expected Result Purpose

10000 3000 Normal case

10000 12000 Boundary / exception case

10000 -500 Invalid input

abc 3000 Invalid input

10 marks

5/5 answered

Your Code

```

1 import java.util.*;
2 class InsufficientBalanceException extends Exception{
3     public InsufficientBalanceException(String message){
4         super(message);
5     }
6     public static void main(String[] args){
7         Scanner sc=new Scanner(System.in);
8         try{
9             double balance=sc.nextDouble();
10            double withdrawal=sc.nextDouble();
11            if(withdrawal<=0){
12                throw new
13                    IllegalArgumentException("Withdrawal amount must be positive");
14            }
15            if(withdrawal>balance){
16                throw new
17                    InsufficientBalanceException("Insufficient balance");
18            }
19            double remainingBalance=balance-withdrawal;
20            System.out.println("Withdrawal Successful");
21            System.out.println("Remaining Balance: "+remainingBalance);
22        }
23        catch(InsufficientBalanceException e){
24            System.out.println(e.getMessage());
25        }catch(IllegalArgumentException e){
26            System.out.println(e.getMessage());
27        }catch(InputMismatchException e){
28            System.out.println("Invalid input. Please enter a valid number.");
29        }finally{
30            System.out.println("Transaction completed.");
31        }

```

Input

10000
3000

Output

Withdrawal Successful.
Remaining Balance: 7000.0
Transaction completed.

```
32         sc.close();  
33     }  
34 }
```

✔ Submitted

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